of hindrance factors. Hence, o-e, e-o and o-o nuclei have normally much longer spontaneous fission half-lives than their neighboring e-e nuclei.

The ground state energy, E, of a nucleus can be regarded as a sum of the liquid drop model energy (including deformation), E_{LDM}, the pairing correction, δ_{P}, and the shell correction, δ_{S}, to that energy.

$$E = E_{\mathrm{LDM}} + \delta_{\mathrm{P}} + \delta_{\mathrm{S}} \tag{6.64}$$

The use of (6.64), with deformation dependent shell corrections, leads to fission barriers with two maxima, see Figure 6.15. The occurrence of a secondary minimum is consistent with the observation of spontaneous fission isomers. Shell and pairing effects also give rise to long spontaneous fission and α-decay half-lives for nuclides around magic N or Z numbers. Detailed calculations based on advanced theories of nuclear structure lead to the prediction of an area of spheroidal nuclei with increased nuclear stability (in their ground state) around $Z = 114$ and $N = 184$, the so called *superheavy elements*, see Ch. 14.

A number of very rare decay modes have also been observed where clusters of nucleons heavier than α-particles are emitted, see §5.7. Some of these are difficult to distinguish from highly asymmetrical fission.

6.8. EXERCISES

6.1. The quantum numbers $s = ½$ and $l = 2$ are assigned to a particle. (a) If spin and orbital movements are independent, how many space orientations (and thus measured spectral lines if no degeneration of energy states occur) are possible in an external field of such a strength that both movements are affected? (b) How many lines would be observed if spin and orbital movements are coupled?

6.2. In the hydrogen atom the K-electron radius is assumed to be 0.529×10^{-10} m (the Bohr radius). (a) Calculate the orbital velocity of the electron assuming its mass to be $\mathbf{m}_{\mathrm{e}}$. (b) How much larger is its real mass because of the velocity? Does this affect the calculations in (a)?

6.3. A beam of protons pass through a homogeneous magnetic field of 0.5 T. In the beam there is a small high frequency coil which can act on the main field so that the proton spin flips into the opposite direction. At which frequency would this occur?

6.4. Calculate the nuclear Landé factor for ^{11}B.

6.5. In Table 6.5 the first degenerate levels have been given. Using the same assumptions, what states will be contained in the next level and how many nucleons will it contain?

6.6. How deep is the nuclear well for ^{116}Sn if the binding energy of the last nucleon is 9 MeV?

6.7. Calculate the spins and nuclear g factors for (a) ^{45}Ca, (b) ^{60}Co, and (c) ^{141}Pr, using data in Table 6.3.

6.8. The observed quadrupole moment of ^{59}Co is 0.40 barn. (a) What is the deformation value β? (b) What spin value is expected from the Nilsson diagram?

6.9. Which neutron and proton states account for the spin value I of ^{14}N?

6.10. A γ-line at 0.146 MeV is assigned to a $+4 \rightarrow +0$ rotational level change in ^{238}Pu. (a) What should the energy of the +2 and +6 rotational levels be? Compare with the measured values of 0.044 and 0.304 MeV. (b) If ^{238}Pu is considered to be a homogenous sphere, what will its apparent radius be? Compare with that obtained using (4.14).

6.11. A ^{239}Pu compound is placed in a test tube in a 40 MHz nmr machine. At what field strength does resonance occur with the nuclear spin? Is the measurement possible? Relevant data appear in Table 6.3.

6.12. Using the Gamow theory the probability for tunneling of an α-particle in the decay of ^{238}U is $1{:}10^{38}$, and the α-particle hits the walls about 10^{21} times per second. What average lifetime can be predicted for ^{238}U from this information?

6.13. Calculate the half-life for α-decay of ^{147}Sm assuming that Q_α is 2.314 MeV. Compare the result with the measured half-life and compute the hindrance factor.

6.9. LITERATURE

I. Perlman, A. Ghiorso, and G. T. Seaborg, Systematics of alpha-radioactivity, *Phys. Rev.* 77 (1950) 26.
I. Perlman and J. O. Rasmussen, Alpha radioactivity, *Handbuch der Physik* 42, Springer-Verlag, 1957.
W. J. Moore, *Physical Chemistry*, 3rd edn., Prentice-Hall, I. 1962.
W. D. Myers and W. J. Swiatecki, Nuclear Masses and Deformations, *Nuclear Physics* **81** (1966) 1.
E. K. Hyde, Nuclear models, *Chemistry* 40 (1967) 12.
H. C. Britt, in N. M. Edelstein (Ed.), *Actinides in Perspective*, Pergamon, 1982, p. 245.
K. S. Krane, *Introductory Nuclear Physics*, Wiley, 1988.
G. T. Seaborg and W. D. Loveland, *The Elements Beyond Uranium*, Wiley-Interscience, 1990.
D. N. Poenaru (Ed.), *Handbook of Decay Modes*, CRC Press, 1993.